College of Computer Sciences and Information Technology

King Faisal University

Fundamentals of Programming

Final Project

Project title:

sport area reservations system

Members:

Atekah hussain aljafar - 222427138

Alaa hussain harabh-222421527

Rawan abdulaziz alsafi-222414494

Submitted to:

Dr.Sharmila banu sheilk Imam

Date:

Feb 2023

I. PROJECT DESCRIPTION

This project is a sports area registration management system that was created to help clubs manage customers' registration, adding, deleting, and modifying materials in an easy way and helping to save time and effort. This system allows storing and accessing all customer's data in one place and collecting all these elements and creating this system is our goal in this project.

II. PROGRAM FEATURES

• Add customer: this feature allows the user to add a new customer by adding the customer's name, phone, and ID.

• Remove customers: this feature allows users to remove added customers by searching for customers added before and deleting them from the system.

• Search customer: this feature searches a customer in the system by comparing every customer ID with the ID entered.

• List customers: this feature will list every customer assigned in the system.

• Update customer: this feature allows the user to update customers' old information with new info that the user enters.

• Add area: this feature allows the user to add a new sports area by completing area info.

• remove area: this feature allows customers to remove reserved areas by their number(ID

• search area: this feature allows customers to search for already existed or added areas by their ID

• list area: this feature will list all areas in the system

• update area: this feature allows customers to change some information or change the reserved areas

• add reservation: this feature asks the user to enter name, phone, and ID and save this information

• remove reservation: this feature deletes information for customers who want to drop out.

II. USER MANUAL

1-The first process that the program performs is adding a customer (customer name, phone, and id)and storing it in the records.

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
1
***************** Add New Customer *****************
 please fill Customer information:
Customer First Name:ateka
Customer Middle Name:
hussain
Customer Last Name:
aljafer
Customer phone:
541310442
Customer ID:
222427138
Customer ID : 222427138|| the Customer: ateka hussain aljafer , || Phone :
    541310442 , || Index Cust : 222427138
added successfully.
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
1
***************** Add New Customer *****************
 please fill Customer information:
Customer First Name:
rawan
Customer Middle Name:
ali
Customer last Name:
mohamed
Customer phone:
503277260
Customer ID:
222414494
Customer ID : 222414494|| the Customer: rawan ali mohamed , || Phone :  503277260
    , || Index Cust : 222414494
added successfully.
************ Welcome to Sport Area Reservation System ****************
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
1
***************** Add New Customer *****************
 please fill Customer information:
Customer First Name:
alaa
Customer Middle Name:
kalid
Customer Last Name:
taher
Customer phone:
541154330
Customer ID:
222421527
Customer ID : 222421527|| the Customer: alaa kalid taher , || Phone :  541154330
    , || Index Cust : 222421527
added successfully.
```

2- The second step is adding a new sports area(game type, players number, price , sports area id)

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
2
***************** Add New Area *****************
 please fill New Area information:
game type:
climbing
players number:
3
price:
60.50
sport area ID:
0
the sport Area: 0 , climbing , 3 By < 60.5 $ added successfully.
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
2
***************** Add New Area *****************
 please fill New Area information:
game type:
swimming
players number:
4
price:
50.50
sport area ID:
1
the sport Area: 1 , swimming , 4 By < 50.5 $ added successfully.
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
2
***************** Add New Area *****************
 please fill New Area information:
game type:
golf
players number:
2
price:
40.50
sport area ID:
2
the sport Area: 2 , golf , 2 By < 40.5 $ added successfully.
```

3-The third step here customer can join to sports area(customer id, sports area id).

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
3
***************** Add New Reservation *****************
 please fill Reservation information:
Customer ID:
222427138
Sport Area ID
1
Customer < ateka hussain aljafer > has Reserved sport area number < 1 > for <
    swimming > game, < 4 > players, by < 50.5 $>.
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
3
***************** Add New Reservation *****************
 please fill Reservation information:
Customer ID:
222421527
Sport Area ID
2
Customer < alaa kalid taher > has Reserved sport area number < 2 > for < golf >
    game, < 2 > players, by < 40.5 $>.
```

```
************ Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
3
***************** Add New Reservation *****************
 please fill Reservation information:
Customer ID:
222414494
Sport Area ID
2
Customer < rawan ali mohamed > has Reserved sport area number < 2 > for < golf >
    game, < 2 > players, by < 40.5 $>.
```

4-The fourth step is the process of deleting a customer reserve when entering(customer id).

```
************ Welcome to Sport Area Reservation System *****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
4
****************** Delete Reservation ******************
 please Enter the ID of Customer you want to delete his Reservation :
222421527
the Reservation of Customer 222421527 and sport Area 1 deleted successfully.
```

5-The fifth step is to delete a customer completely when entering (id of the customer).

```
************ Welcome to Sport Area Reservation System *****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
5
****************** Delete Customer ******************
 please Enter the ID of Customer you want to delete :
222421527
the Customer deleted successfully.
```

6-The sixth step is the process of deleting the sports area that has been added by entering the id of the sports area.

```
************ Welcome to Sport Area
     Reservation System *****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*************************************************
    *********
Enter your choice:
6
******************* Delete Sport Area
    *******************
 please Enter the ID of Sport Area you want to
     delete :
2
the Sport Area deleted successfully.
```

7-The seventh step is a search process, searching for the customer by the customer id, when the customer id enters the program will search if the customer exited or not.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
7
***************** Search Customer *****************
 please Enter the ID of Customer you want to find :
222414494
yes .. the Customer is existed.
```

8-The eighth step is a search process, searching for the sports area by the sport id, when the sport id enters the program will search if the sports area exists or not.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
8
***************** Search Sport Area *****************
 please Enter the ID of Sport Area you want to find :
1
yes .. the Sport Area is existed.
```

9-The ninth step is to display the list of customers who have been assigned.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
9
****************** List Of Customers ******************
  ID  |      Name      |   Phone
-------------------------------------------------------------
222427138 ateka         541310442
222414494 rawan         503277269
```

10-The tenth step of the program displays a list of sports areas that have been added.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
10
***************** List Of Sport Areas  *****************
  ID  |      Game      |   Players   |   Price
-------------------------------------------------------------
0      climbing      3      60.5
1      swimming      4      50.5
```

11-The eleventh step is the modification process, it is the entry of the customer id that will modify the customer name and phone.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
11
***************** Update Customers *****************
 please Enter the ID of Customer you want to update :
222414494
please fill New information:
Customer New First Name
RAWAN
Customer New middle Name
ALI
Customer New last Name
MOHAMMED
Customer New Phone
541222548
the Customer: 222414494 , RAWAN ALI MOHAMMED , 541222548 updated succes
```

12-The twelfth step is the modification process of sport area details when entering the sports area id, can modify the sports area name, number of players, and price.

```
 *********** Welcome to Sport Area Reservation System ****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
12
***************** Update Sport Areas *****************
 please Enter the ID of Sport Area you want to update :
0
please fill New information:
Sport Area New Game
CLIMBING
Sport Area New Players
6
Sport Area New Price
130
the sport Area: 0 , CLIMBING , 6 updated successfully.
```

13-The thirteenth step is the step to exit the program permanently.

The program will repeat itself until the exit process is completed.

```
************ Welcome to Sport Area Reservation System *****************
Choose what do you want to do:
1.  Add new Customer.
2.  Add new Sport Area.
3.  Reserve sport area by a Customer.
4.  Remove Customer reservation.
5.  Delete a Customer.
6.  Delete Sport Area.
7.  Search for Customer.
8.  Search for Sport Area.
9.  List the Customers.
10. List the Sport Areas.
11. Update Customer Details.
12. Update Sport Area Details.
13. Exit.
*****************************************************
Enter your choice:
13
```

IV. Program Code

fop progect final.cpp